Characteristics of Dashboard Transcript

Let us look at the characteristics of dashboards. There are seven characteristics of dashboards and each of them will help to understand how to effectively design a dashboard. First is to understand that dashboard is nothing but condensation of a complex data into visual summaries.

This is visual summarization. What is a dashboard? As I said, it's a collection of a group of visuals. It is condensing the complex information about a process, let us say, a website uptime.

So there are how much is the traffic flow, which channels is it coming, what is the bandwidth that is being used, are there any attacks, are there any downtime reports that has been raised. So all these things would be complex data streams put together to understand, to give the feel to the operator that how is the website is functioning. Likewise, you have a single screen display for the effectiveness.

If you recall, your dashboard has to be small, concise, clear, right, and it will be mostly without scrolling or clicking because you need to get the insights quickly. And there might be some level of functionality added, so that you can immediately get the insights if you want finer details. And it is almost real time or near real time.

Especially from an operations dashboard standpoint, you need to see the data in real time to take actions or near real time with very less lag. There may be a few minutes to a few seconds that is fine if depending upon the operations or you are doing end of day sales analysis, a day is also fine. But this again depends on the business context in which you are operating.

Finally, the customisation and interactivity as you go the decisioning ladder or you are an analyst or a strategic person, you want to know the reasons. So there will be interactivity for you to understand what are the reasons behind the visual insights. So this is what the narrative driven by the reader looks like.

It tries to work with the insights that the ready visuals are providing, but it would also make the audience or the reader fetch some questions and use the interactivity on the dashboard to answer those questions. Finally, you have contextual relevance, wherein you try to answer a particular business question to the audience using the visuals. And you have to use the pre attentive attributes that we have learned in the earlier classes to design effective visuals.

We have seen all the pre attentive attributes to design effective visuals in individual level. Now we are bringing them together on a dashboard together these principles should also hold together so that the overall visual looks effective and expressive. So again, let us talk about the characteristics of dashboard using an example.

This is real GDP growth rate from again International Monetary Fund and this is I have specifically chosen the year 2020 in which COVID has struck the world. So the dashboard clearly shows let me first walk you through the elements of this dashboard. The dashboard clearly shows the growth differentials in the world countries, you can see the ones who are in red had less than 3% growth or negative 3% growth.

The ones who had in green were about three light green is 0 to 3, dark green is 3 to 6 and even more dark green is greater than 6%. And there are certain countries where there is no data. So clearly the colour coding is visually striking.

What is the immediate conclusion you can draw when you look see you can assume that the entire countries in the world the majority countries in the world have lesser growth rate than compared to general time and there is a interactive time player where you can seek the timeline and understand how this has been varying over a period of time. Then again in interactivity, you can choose a particular country or a region or an analytical group. In this case analytical group would be low income countries, middle income countries or advanced economies or Asia and Southeast Asia like that.

So once you look at this dashboard is all already doing a visual summarization of the complex data, what is the complex data, the data that you are seeing here, which is nothing but countrywide growth rate is really complex. You have 183 countries how to summarise them effectively, this visual using the spatial map and the colour saturation, which is one of the preattentive attributes and grading them on different colour hues or different darkness of colours has intuitively shown that higher growth rates with dark green and lighter growth rates or negative growth rates lesser growth rates with red giving a visual appeal that growth rate has fallen during the COVID-19 pandemic. Now again look at the same graph for the 2024 period, you can see the entire graph becoming green clearly showing that world as a whole has seen far more better growth rate as compared to 2020 period.

Now what we have to understand here is if you are highlighting on a particular country you are getting a small pop-up saying that India at 6.5 is a fastest growing country and you have all these elements that are embedded into the graph so that the interactivity is also that I want to understand India I can just take on this or go to the region and see how India is comparing it to its peers and then I can use the timeline to see how over a period of time these growth rates have changed. This kind of interactivity enables a reader-driven narrative to absorb the context of how growth rate is moving over the countries. So again this dashboard almost all captures the key seven characteristics of the dashboard.

Now let us take one more example in this dashboard the aim is to show the frequency of natural disasters and this has been plotted over a period of time starting from 1995 have cut and shown the recent time period and if you see it is given at the country level there is an interactivity so you can just select the drop down and you can select a country I have chosen India and I have deselected all others and taken only flood and on top of this when I move the cursor I get a pop-up again saying that in 2006 there are 17 flood related natural disasters in India. So it clearly shows that the length of the bar when you are looking at the graphic you are the length of the bar reflects the physical quantity and to help you to understand that better when you move the cursor it is also showing a pop-up. So it is clearly capturing the visual attribute space while looking at the graphic itself you can get some sense that the flood intensity or the flood frequency of natural disasters because of floods has fallen over a period of time in India as compared to 2000-2005 pre-2005.

Now I have selected all other types of natural disasters and we can see in the recent period the storm related natural disasters are on the increase as compared to other natural disasters and landslides are also increasing. You can clearly get certain views and then whenever you have this pop-up element this is an interactive feature where the author is helping the audience to get clear insights when they are moving. Imagine if there is no pop-up function you have to physically compute the 11 number or the get an estimate about what is the length of the bar indicating.

Remember when you decode any graph you should be clearly able to convey the physical information or when you decode any graph your graphic should be able to intuitively tell the estimation that it is trying to represent. So in this way by using pop-ups in the dashboard you are able to create a very quick you are able to give a quick insight to your audience. We are now pausing for a quiz.

Thank you.

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